

TAYLOR PENTZ

Senior Software Engineer

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Professional Summary

Innovative and results-driven Senior Software Engineer with over 10 years of hands-on experience across backend systems, distributed architecture, and blockchain-integrated apps. Proven ability to build and scale production-ready APIs, contribute to smart contract development, and lead full-stack features from design to deployment. Passionate about Web3, DeFi, and crafting resilient systems using modern JavaScript/TypeScript, Node.js, PostgreSQL, Redis, and cloud-native tooling.

Technical Skills

- ❖ **Languages:** JavaScript, TypeScript, Solidity, SQL, Python, Rust (familiar)
- ❖ **Frontend:** React, Next.js, Redux, Tailwind, Web3.js, Ethers.js
- ❖ **Backend:** Node.js, Express.js, NestJS, RESTful APIs, Redis, PostgreSQL, MongoDB
- ❖ **Blockchain:** Smart Contracts (Solidity), EVM, dApps, DAO tools, Ethereum, Layer 2 scaling
- ❖ **Infrastructure:** Docker, GitHub Actions, CI/CD pipelines, AWS, GCP, Azure
- ❖ **Testing & DevOps:** Jest, Mocha, Cypress, Postman, Swagger, Git, Agile, TDD
- ❖ **Database:** PostgreSQL, MongoDB, Redis, MySQL, NoSQL, SQL optimization

Professional Experience

SumatoSoft

Senior Software Engineer

Boston, MA (Remote) | Apr 2023 – Mar 2025

- Architected and deployed scalable RESTful APIs for decentralized financial platforms, ensuring sub-200ms latency and secure on-chain data access using Node.js and PostgreSQL.
- Built event-driven data pipelines leveraging Redis for high-throughput caching and smart contract indexing, processing 2M+ daily transactions.
- Contributed to Solidity-based smart contract development, including testnets deployment and collaborative audit sessions targeting reentrancy, overflow, and gas optimizations.
- Delivered end-to-end full-stack features with React and TypeScript, integrating EVM-based smart contracts via Web3.js and Ethers.js.
- Implemented Dockerized microservices orchestrated via GitHub Actions for CI/CD, improving deployment frequency by 40%.
- Led integration of secure wallet connections using WalletConnect and MetaMask to support multi-chain onboarding and DAO voting interfaces.
- Mentored 3 junior developers in blockchain application architecture, Solidity testing patterns, and secure dApp deployment.
- Conducted performance profiling and debugging of real-time transaction feeds for multi-chain compatibility across Ethereum and Layer 2 rollups.

Hexaview Technologies Inc.

Software Engineer

New York, NY | Jul 2018 – Mar 2023

- Engineered multi-tenant back-end services for DeFi dashboard platforms using Express.js, ensuring SLA-compliant API uptime of 99.95%.
- Designed PostgreSQL schemas for blockchain event tracking, enabling 5x faster query execution through optimized indexing and partitioning.
- Integrated smart contract events via WebSockets and JSON-RPC for Solana and Ethereum, syncing contract states in near real-time.
- Developed internal SDKs to abstract smart contract interactions and improved gas cost tracking across dApp modules.
- Implemented authentication via JWT and role-based access control, securing API endpoints and reducing attack surfaces.
- Built server-side cron jobs for NFT metadata indexing and collection-level analytics syncing with IPFS/Arweave.
- Participated in code reviews and security evaluations, contributing to safe deployment pipelines for EVM-compatible dApps.
- Collaborated with product and smart contract teams to launch governance features and liquidity pools across testnets and mainnet.

SimbirSoft Company

Software Developer

Boston, MA | Mar 2015 – Jun 2018

- Created modular APIs and admin dashboards for financial SaaS applications, leveraging PostgreSQL and Redis for scalable data workflows.
- Delivered dynamic interfaces in React and TypeScript, including real-time portfolio monitoring integrated with blockchain oracles.
- Developed internal tools to visualize staking metrics, gas fees, and liquidity volumes across Web3 ecosystems.
- Refactored monolithic services into Dockerized containers, enabling seamless CI/CD adoption and platform modernization.
- Participated in smart contract development bootcamps and conducted PoCs for internal DeFi strategies including flash loan simulations.
- Streamlined front-end component reuse by implementing a shared UI system based on React hooks and state containers.
- Designed alerting system for validator downtime and token volatility using Grafana and Node.js workers.
- Enhanced code quality through peer reviews and enforced consistent linting, unit testing, and branching strategies.

Microsoft

Software Developer Intern

Cambridge, MA | Aug 2014 – Feb 2015

- Prototyped secure key management microservices for internal blockchain research tools using Node.js and Redis.
- Assisted in developing interactive data visualizations for cryptographic proof structures using D3.js and TypeScript.
- Contributed to Ethereum-related research, writing test harnesses for Geth/Parity clients and measuring block propagation delays.
- Built performance benchmarks comparing various hash functions and transaction validation cycles.
- Supported engineering team during early dApp proof-of-concept initiatives targeting identity and supply chain use cases.
- Automated CI workflows using Jenkins and improved build reliability for blockchain test suites.
- Reviewed academic papers on consensus algorithms and translated concepts into unit-tested algorithms for simulations.
- Participated in weekly design sessions focused on interoperability between smart contracts and off-chain logic.

Projects

DeFi Smart Vault Automation (2023)

- Led development of a vault system that automated rebalancing across AMMs using Solidity, Node.js, and Redis.
- Achieved 30% higher yield efficiency by integrating real-time pricing via Chainlink and on-chain position monitoring.

Layer 2 NFT Bridge (2022)

- Architected a cross-chain NFT bridge between Ethereum and Arbitrum using event-driven architecture and Ethers.js.
- Reduced asset transfer time by 60% while implementing fraud-proof verification and dual-signature authorization.

DAO Governance Dashboard (2021)

- Built a governance portal enabling real-time voting, proposal indexing, and ENS-integrated member dashboards.
- Deployed across Polygon and Optimism testnets, handling 10K+ active wallets securely.

Certifications

- AWS Certified Solutions Architect – Associate
- Google Associate Cloud Engineer
- MongoDB for Developers (MongoDB University)
- Python for Data Structures and Algorithms (Coursera)

Education

University of North Carolina at Chapel Hill

Bachelor's Degree in Computer Science (2010 – 2014)